Vectors in a Plane

section 11.1

8-12-26

A) Quick Review

1. Component Form $\langle v_1, v_2 \rangle$

initial $(3, 8)$ terminal $(2, -5)$

$\langle 2-3, -5-8 \rangle$

$\langle -1, -13 \rangle$

2. Magnitude:

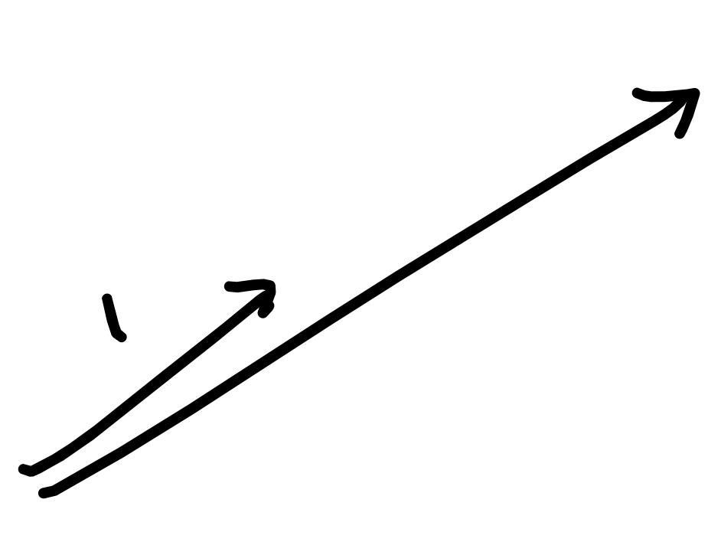
$\|\vec{v}\| = \sqrt{1 + 13^2} = \sqrt{167}$

$\|\vec{v}\| = 1$ unit vector

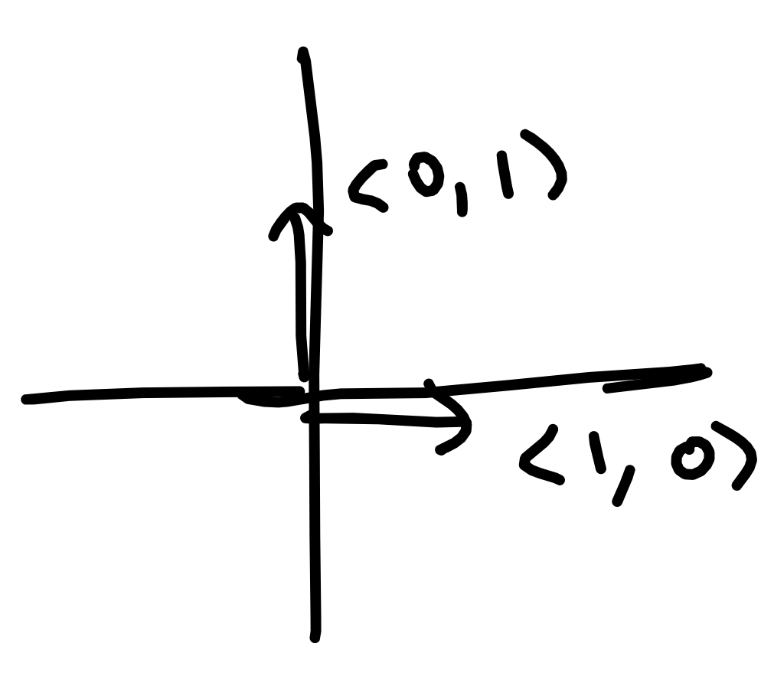
$\|\vec{v}\| = 0$ zero (null) vector

3. Unit Vector in direction $\bar{v}$

$\vec{u} = \frac{1}{\|\vec{v}\|}(\vec{v})$



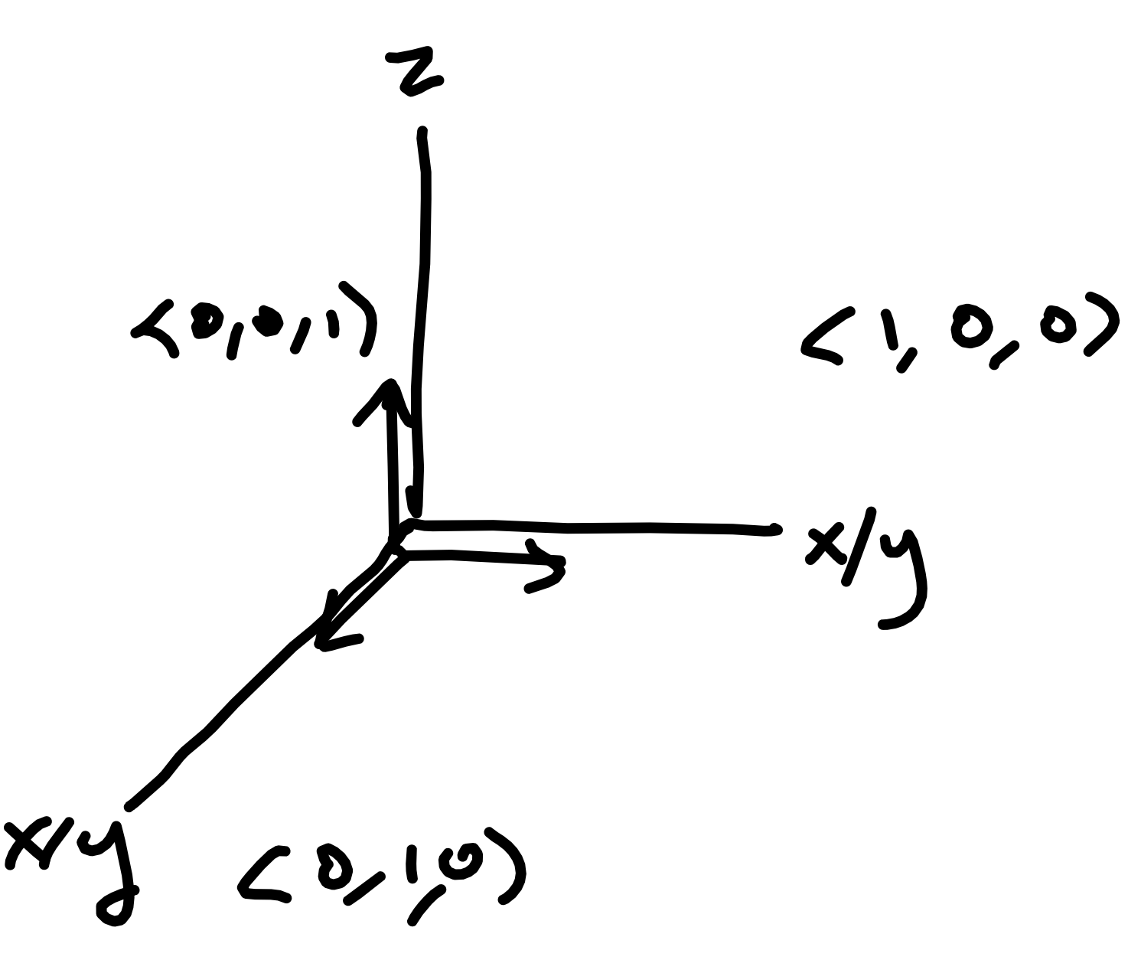
4. Standard Unit Vector



i

j $3i + 2j$

$\langle 3, 2 \rangle$



Ex: Given magnitude 4 and $\vec{u} = \langle 3, -2 \rangle$

$\|\vec{u}\| = \sqrt{9+4} = \sqrt{13}$

$4\left(\frac{1}{\sqrt{13}}(\langle 3, -2 \rangle)\right) = \left\langle \frac{12}{\sqrt{13}}, \frac{-8}{\sqrt{13}} \right\rangle$

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9) $\langle 3, 5 \rangle$ 15) $\langle -1, 1\frac{2}{3} \rangle$

19) a) $\langle \frac{8}{3}, 6 \rangle$

b) $\langle 6, -15 \rangle$

c) $\langle -2, -14 \rangle$

d) $\langle 10, -7 \rangle$

27) terminal $\langle 3, 5 \rangle$

35) $\frac{1}{\sqrt{153}}(\langle 3, 12 \rangle)$

39) a) $\sqrt{2}$ b) $\sqrt{5}$ c) 1

d) 1 e) 1 f) 1

45) $\langle 0, 6 \rangle$

47) $\langle -\sqrt{5}, 2\sqrt{5} \rangle$

B) Component Forms

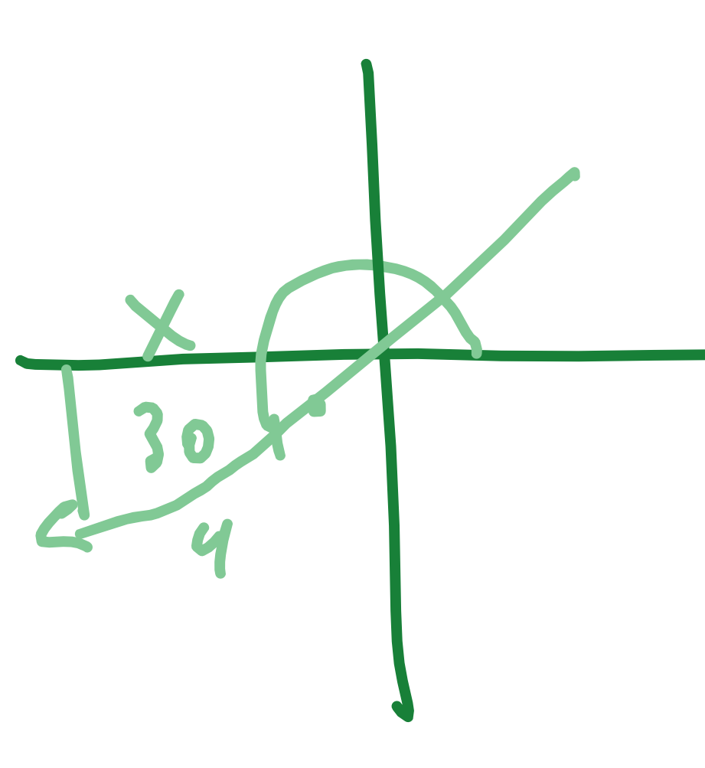
ex. $\|\vec{v}\| = 4$

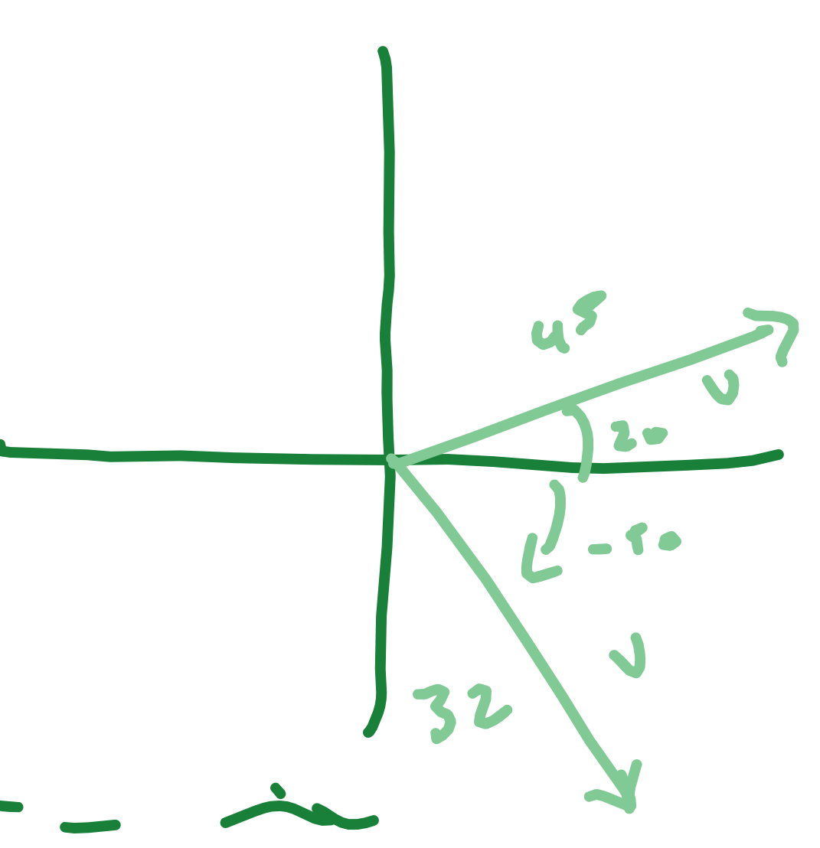
w/ pos x-axis 210°

$\langle r\cos\theta, r\sin\theta \rangle$ (x, y)

$\langle 4\cos 210, 4\sin 210 \rangle$

$\langle -2\sqrt{3}, -2 \rangle$



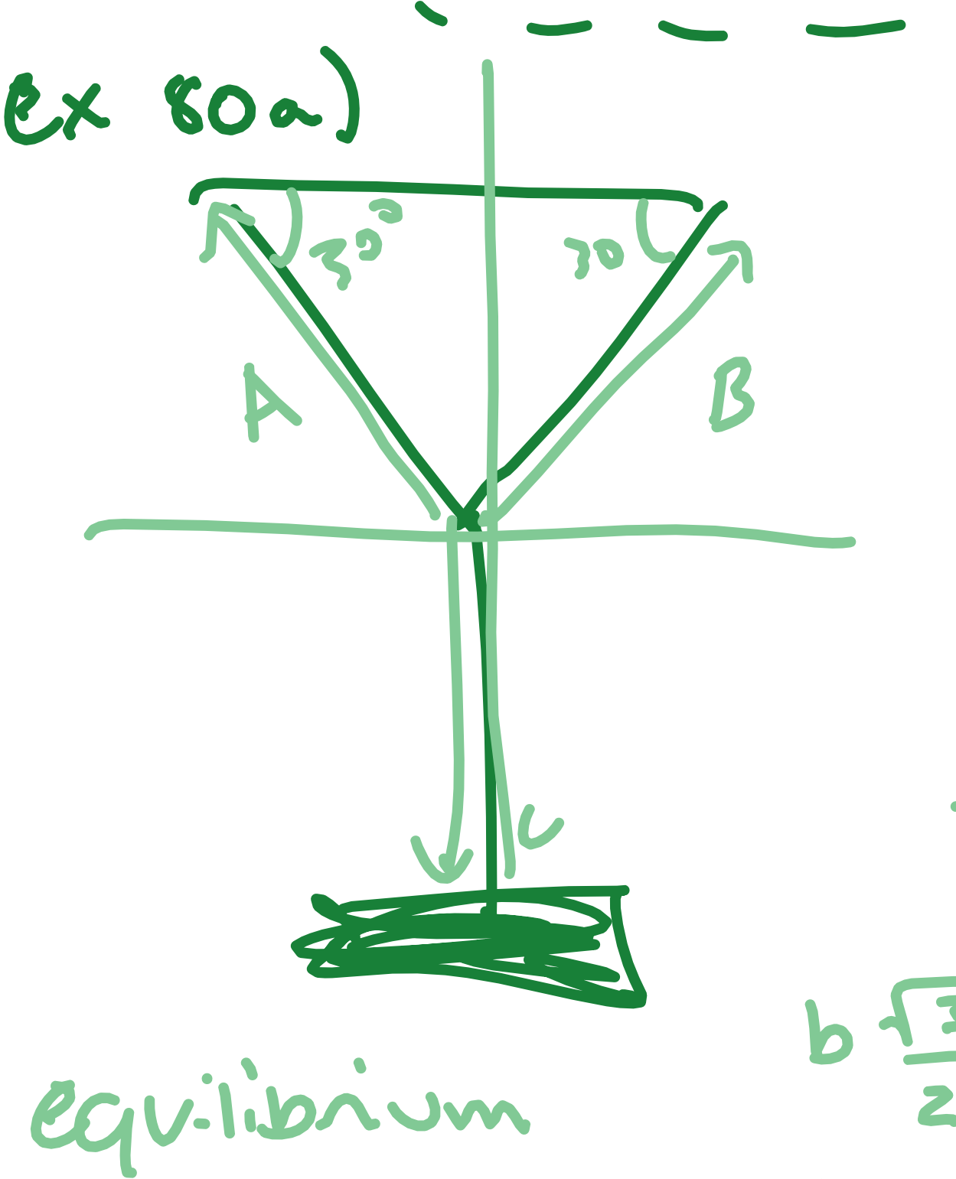


$\vec{u} = \langle 45\cos(20), 45\sin(20) \rangle$

$\vec{v} = \langle 32\cos(50), -32\sin(50) \rangle$

$\vec{u} + \vec{v} = \langle 62.855, -9.123 \rangle$

ex 80a)



$\vec{C} = \langle 0, -3000 \rangle$

$\vec{B} = \langle b\cos 30°, b\sin 30° \rangle$

$= \langle b\frac{\sqrt{3}}{2}, b\frac{1}{2} \rangle$

$\vec{A} = \langle a\cos 130, a\sin 130 \rangle$

equilibrium

so A+B+C=0

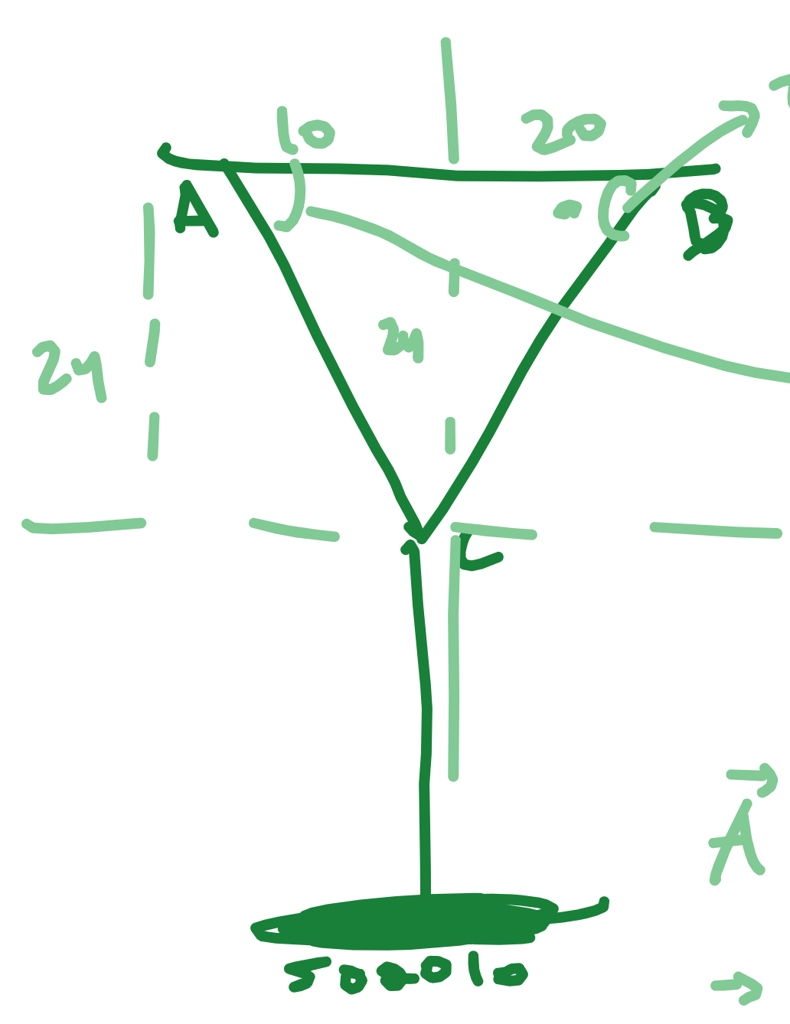
$b\frac{\sqrt{3}}{2} + a\cos 130° = 0$

$\frac{b}{2} + a\sin 130° = 3000$

a = 2638.156

b = 1958.111

ex 80b)



$\tan B = \frac{24}{20}$

$B = 0.876058$

$= \arctan(\frac{24}{20})$

$\tan A = \frac{24}{10}$

$A = \pi - \arctan(\frac{24}{10})$

$= 1.966$

$\vec{A} = \langle a\cos(1.966), a\sin(1.966) \rangle$

$\vec{B} = \langle b\cos(0.876), b\sin(0.876) \rangle$

$\vec{C} = (0, -5000)$

equilibrium

A+B+C=0

$A_x + B_x = 0$

$A_y + B_y = 5000$

$a\cos(1.966) = -b\cos(0.876)$

$-\frac{a\cos(1.966)}{\cos(0.876)} = b$

$b = -0.601$

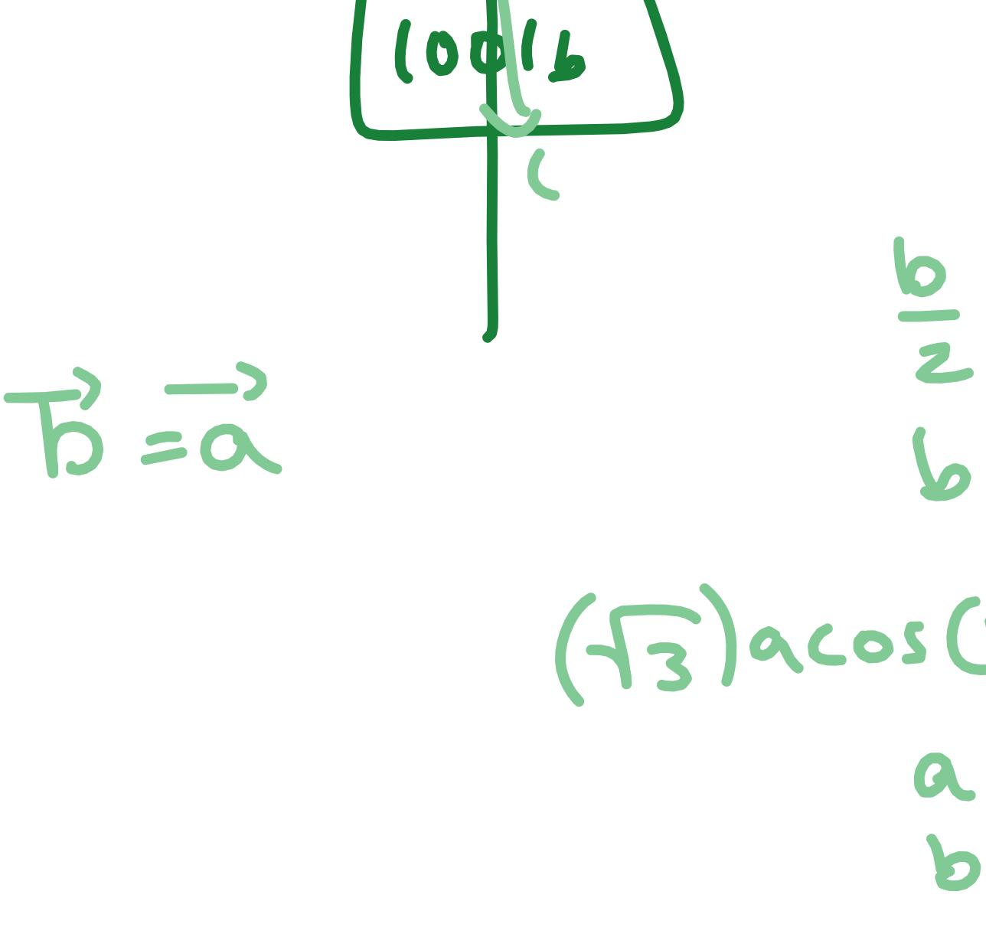
$a\sin(1.966) + b\sin(0.876) = 5000$

$a\sin(1.966) - 0.601a\sin(0.876) = 5000$

a = 3611.12

b = 2169.49

ex 82)



$\vec{c} = \langle 0, -100 \rangle$

$\vec{b} = \langle \frac{b}{2}, \frac{b\sqrt{3}}{2} \rangle$

$\vec{a} = \langle a\cos(110°), a\sin(110°) \rangle$

$\vec{b} = \vec{a}$

$\frac{b}{2} = a\cos(110°)$

$b = 2a\cos(110°)$

$(\sqrt{3})a\cos(110°) + a\sin(110°) = 5000$

a = 3263.52

b = 2232.38